Pohl's Pendulum

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Aim of the Experiment

To determine the damping constant of the pendulum for different eddy damping currents.

Apparatus Required

(i) Pohl's pendulum set up with power supply
(ii) Bridge rectifier
(iii) Stop watch
(iv) Multimeter & connecting wire

Fig-1: Experimental Setup

Theory

Pohl's pendulum is a torsional pendulum which swings back and forth under the action of an external force. The complete experimental set up is shown in Fig-1. When a system oscillates, the amplitude of oscillation decreases with time due to natural damping (when no other force is acting on the pendulum). To keep the pendulum in oscillation mode and maintain the amplitude of the oscillation at a particular value ϕ, energy should be pumped into the system from the external source.

A. <u>Underdamped Oscillation</u>

In a torsional pendulum the restoring torque, M_1, and the damping torque (resistance), M_2 are given by,

$M_1 = -D\phi$ and $M_2 = -C\phi$

Where ϕ= angle of rotation due to the external force imparted on the pendulum.

D= torque per unit angle

C= proportionality factor depending on the strength of current which initiates eddy current. This results in the following equation of motion,

$$I\ddot{\phi} + C\dot{\phi} + D\phi = 0 \quad \text{.....} \quad (1)$$

where, I= moment of inertia of the pendulum about the axis of rotation and $\ddot{\phi}$*= angular acceleration.*

Dividing eqn (1) by *I* we obtain,

$$\ddot{\phi} + 2\beta\dot{\phi} + \omega_0^2\,\phi = 0 \quad \text{.....} \quad (2)$$

where $\beta=\frac{C}{2I}$ *and* $\omega_0^2=\frac{D}{I}$.

β is named as the damping constant and ω_0 is the natural frequency of the undamped system.

If the pendulum is rotated to one side and released from rest at t=0, such that $\phi=\bar{\phi}$ and $\dot{\phi}=0$ at t=0, then the solution of the equation (2) is

$$\phi(t) = \phi_0 e^{-\beta t} \cos(\omega t+\delta) \quad \text{.....(3)}$$

$$\phi_0=\bar{\phi}\sqrt{1+\frac{\beta^2}{\omega^2}} \text{ and } \delta=\tan^{-1}\left(-\frac{\beta}{\omega}\right) \text{ with } \omega=\sqrt{\omega_0^2-\beta^2}\geq 0$$

This shows that the amplitude of underdamped oscillations decreases exponentially with time.

The constant λ is obtained from the log of the ratio of the successive amplitudes, which is expressed mathematically as,

$$\lambda = \ln\left(\frac{\phi_n}{\phi_{n+1}}\right) = \beta T$$

We have $\frac{\phi_n}{\phi_{n+1}} = \frac{\phi_0\, e^{\beta t} Cos\,(\omega t+\delta)}{\phi_0\, e^{\beta(t+T)} Cos\,(\omega t+\delta)}$

So $\lambda = \ln e^{\beta T} = \beta T$, where $T = 2\pi/\omega$ is the time period.

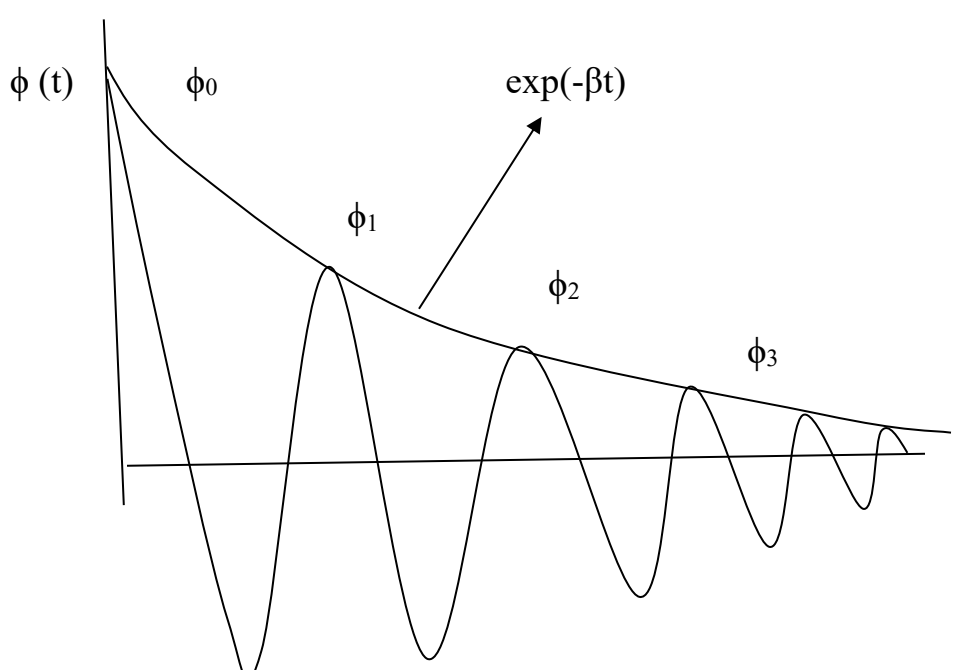

Fig-2: Underdamped oscillation.

Case -1

For $\omega_0^2 = \beta^2$, the pendulum returns to its initial position in minimum time without oscillation. This is the condition for critically damped oscillation

Case-2

For $\omega_0^2 < \beta^2$, the pendulum returns to its initial position asymptotically. This is called the condition of creeping or over damped oscillation.

B. Forced oscillation

For forced oscillation an external torque $M_a = M_0 \, Cos \, \omega_a t$, is applied to the pendulum. Thus we get the new equation of motion as,

$$\ddot{\phi} + 2\beta\dot{\phi} + \omega_0^2\phi = F_0 \, Cos \, \omega_a t \quad \text{---------- (4)}$$

Where $F_0 = \frac{M_0}{I}$

The steady state solution of this differential equation gives

$$\phi(t) = \phi_a \, Cos \, (\omega_a - \alpha) \quad \text{----------(5)}$$

Where $\phi_a = \frac{F_0}{\sqrt{(\omega_0^2 - \omega_a)^2 + 4\beta^2\omega_a^2}}$ ---------- (6)

and $\tan \alpha = \frac{2\beta\omega_a}{(\omega_0^2 - \omega_a^2)}$ ----------(7)

Looking at Eq. 6 we observe that,

(i) *When* F_0 *increases* ϕ_a *also increases.*

(ii) *For a fixed value of* F_0, ϕ_a *has a peak value and the corresponding frequency is the resonance frequency given by* $\omega_{res} = \sqrt{(\omega_0^2 - 2\beta^2)} = \sqrt{(\omega^2 - \beta^2)}$.

(iii) *The greater the value of* β, *the smaller is* ϕ_a.

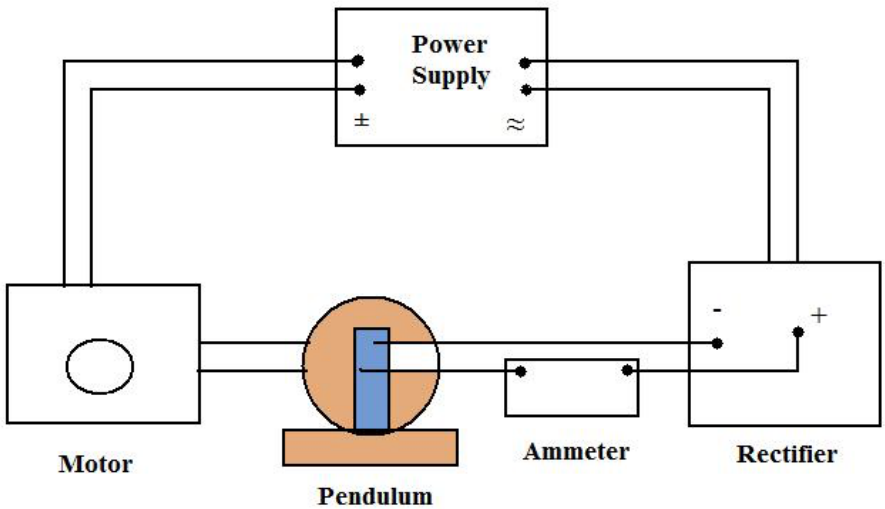

Fig-3: Electrical connections of the set up

Procedure

<u>*Underdamped Oscillation*</u>

(i) The pointer of the pendulum was deflected to one side (the pointer was brought to 19 of the circular scale) taking care that the pendulum does not cross the wire. Then it was let free and the pendulum was set for oscillation. Three independent observations were taken for 20 oscillations. The time period of oscillation was found from the average.

(ii) Successive unidirectional amplitudes were measured(on the other side of initial deflection) of oscillation after every time interval T (i.e. ϕ_1 at t_1 ,ϕ_2 at t_2.....φ_{10} at t_{10}).

(iii) Now the system was connected to power source. The current value was set to i_1. The time period was found out according to the procedure described in (i). Then repeated (ii) for $i=i_1$. The corresponding voltages were noted.

(iv) The procedure given in (i), (ii),(iii) for $i=i_2$, $i=i_3$ was repeated with the same initial deflection to minimize errors in the measurements of amplitudes. The corresponding voltage for $i=i_2$ & $i=i_3$ were noted.

(v) Successive unidirectional amplitudes as a function of time for different damping currents are plotted in a semi log graph paper (time on the normal axis and amplitude on log axis). For a fixed damping plot should be a straight line. From the slopes of the plots, damping resistance, β *can be estimated for various damping.*

<u>*Forced Oscillation*</u>

(i) The motor was switched on and kept the coarse and fine knob of the motor at low and medium position respectively. The damping voltage was kept at 2v.

(ii) The forcing frequency of the motor ω_a can be found out by counting the number of turns the disc makes per unit time with a stop watch.

(iii) The frequency of the motor was slowly increased and the amplitude of the pendulum was measured when it had stabilised. The frequency of the motor is increased in small steps turning the fine knob.

(iv) The maximum amplitude or the resonance is attained and the resonance frequency $\omega_a = \omega_{res}$ is measured.

(v) For small value of damping, care must be taken such that the forcing frequency should not exceed certain limit and the pendulum does not cross the scale at resonance.

(vi) The experiment is repeated for different values of damping current. Plot separate graph between the forcing frequency and the amplitude for different value of damping current.

(vii) The resonance frequency was found from the graph and compared with the result that obtained theoretically.

Observations

Table-1: Time period for i=0 amp.

Sl. No.	Time for 20 oscillations in sec	Time Period (T) T= **(t_1+ t_2+ t_3)/60 in sec**
1	**t_1= 37.96**	**1.88**
2	**t_2= 37.34**	
3	**t_3= 37.83**	

Table-2: Amplitude for i=0 amp.

Time (in sec)	Amplitude		$\phi_{avg} = (\phi_{ia}+\phi_{ib})/2$
	a	b	i=1 to 10
t_1=T/2=0.94	18.2 =ϕ_{1a}	19.2 =ϕ_{1b}	$\phi_1 = (\phi_{1a}+\phi_{1b})/2$=18.7
t_2=3T/2=2.82	17.9 =ϕ_{2a}	18.8 =ϕ_{2b}	$\phi_2 = (\phi_{2a}+\phi_{2b})/2$=18.4
T_3=5T/2=4.70	17.6 =ϕ_{3a}	18.6 =ϕ_{3b}	$\phi_3 = (\phi_{3a}+\phi_{3b})/2$=18.1
t_4=7T/2=6.58	17.4 =ϕ_{4a}	18.2 =ϕ_{4b}	$\phi_4 = (_{34a}+\phi_{4b})/2$=17.8
t_5=9T/2=8.46	16.4 =ϕ_{5a}	17.8 =ϕ_{5b}	$\phi_5 = (\phi_{5a}+\phi_{5b})/2$=17.1
t_7=11T/2=10.34	15.8 =ϕ_{6a}	17.6 =ϕ_{6b}	$\phi_6 = (\phi_{6a}+\phi_{6b})/2$=16.7
t_7=13T/2=12.22	15.4 =ϕ_{7a}	17.0 =ϕ_{7b}	$\phi_7 = (\phi_{7a}+\phi_{7b})/2$=16.2
t_8=15T/2=14.1	14.6 =ϕ_{8a}	16.8 =ϕ_{8b}	$\phi_8 = (\phi_{8a}+\phi_{8b})/2$=15.7
t_9=17T/2=15.98	14.2 =ϕ_{9a}	16.2 =ϕ_{9b}	$\phi_9 = (\phi_{9a}+\phi_{9b})/2$=15.2
t_{10}=19T/2=17.86	13.8 =ϕ_{10a}	15.8 =ϕ_{10b}	$\phi_{10} = (\phi_{10a}+\phi_{10b})/2$=14.8

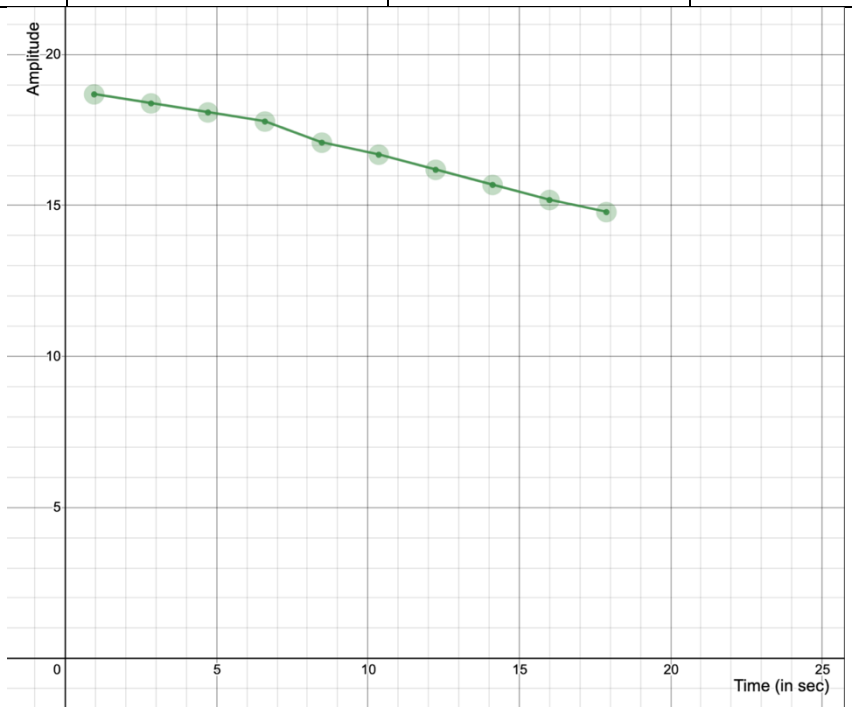

Table-3: Time period for i=0.2 amp.

Sl. No.	Time for 20 oscillations in sec	Time Period (T) T= **(t_1+ t_2+ t_3)/60 in sec**
1	**t_1= 37.08**	**1.87**
2	**t_2= 37.63**	
3	**t_3= 37.49**	

Table-4: Amplitude for i=0.2 amp.

Time (in sec)	Amplitude		$\phi_{avg} = (\phi_{ia}+\phi_{ib})/2$ i=1 to 10
	a	b	
t_1=T/2=0.94	17.8 =ϕ_{1a}	18.2=ϕ_{1b}	$\phi_1 = (\phi_{1a}+\phi_{1b})/2$=18
t_2=3T/2=2.80	16.2 =ϕ_{2a}	17.0 =ϕ_{2b}	$\phi_2 = (\phi_{2a}+\phi_{2b})/2$=16.6
T_3=5T/2=4.68	14.2 =ϕ_{3a}	15.4 =ϕ_{3b}	$\phi_3 = (\phi_{3a}+\phi_{3b})/2$=14.8
t_4=7T/2=6.54	12.2 =ϕ_{4a}	13.8 =ϕ_{4b}	$\phi_4 = (_{34a}+\phi_{4b})/2$=13
t_5=9T/2=8.42	11.0 =ϕ_{5a}	12.2 =ϕ_{5b}	$\phi_5 = (\phi_{5a}+\phi_{5b})/2$=11.6
t_7=11T/2=10.28	9.8 =ϕ_{6a}	11.0 =ϕ_{6b}	$\phi_6 = (\phi_{6a}+\phi_{6b})/2$=10.4
t_7=13T/2=12.16	8.8 =ϕ_{7a}	9.6 =ϕ_{7b}	$\phi_7 = (\phi_{7a}+\phi_{7b})/2$=9.2
t_8=15T/2=14.02	7.8 =ϕ_{8a}	8.8 =ϕ_{8b}	$\phi_8 = (\phi_{8a}+\phi_{8b})/2$=8.3
t_9=17T/2=15.90	6.8 =ϕ_{9a}	7.8 =ϕ_{9b}	$\phi_9 = (\phi_{9a}+\phi_{9b})/2$=7.3
t_{10}=19T/2=17.76	6.0 =ϕ_{10a}	6.6 =ϕ_{10b}	$\phi_{10} = (\phi_{10a}+\phi_{10b})/2$=6.3

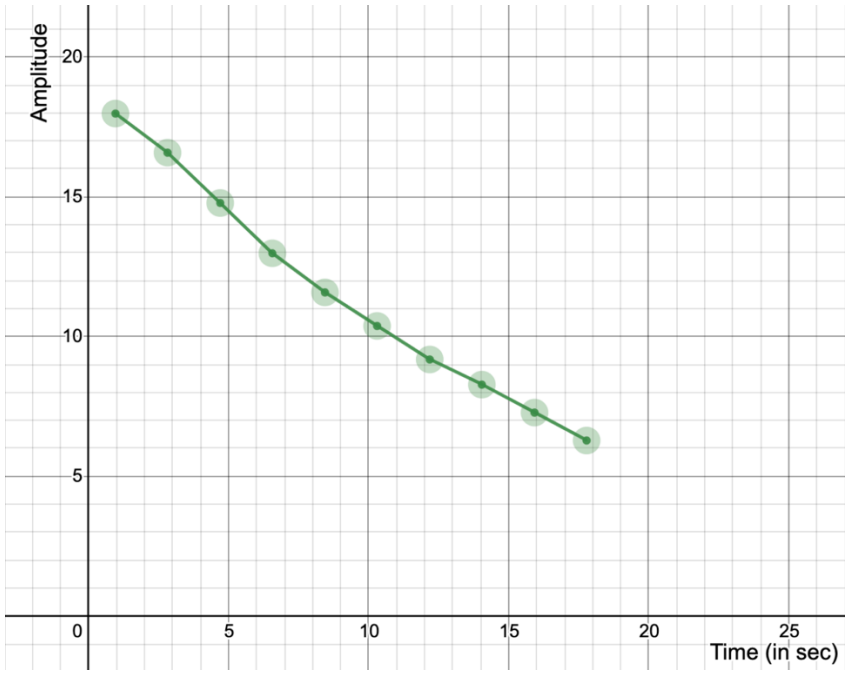

Table-5: Time period for i=0.4 amp.

Sl. No.	Time for 10 oscillations in sec	Time Period (T) T= **(t_1+ t_2+ t_3)/30 in sec**
1	**t_1= 18.59**	**1.86**
2	**t_2= 18.74**	
3	**t_3= 18.32**	

Table-6: Amplitude for i=0.4 amp.

Time (in sec)	Amplitude		$\phi_{avg} = (\phi_{ia}+\phi_{ib})/2$ i=1 to 9
	a	b	
t_1=T/2=0.93	16.2 =ϕ_{1a}	15.8 =ϕ_{1b}	$\phi_1 = (\phi_{1a}+\phi_{1b})/2$=16

t_2=3T/2=2.79	10.2 =ϕ_{2a}	9.8 =ϕ_{2b}	$\phi_2 = (\phi_{2a}+\phi_{2b})/2$=10
T_3=5T/2=4.65	6.0 =ϕ_{3a}	6.8 =ϕ_{3b}	$\phi_3 = (\phi_{3a}+\phi_{3b})/2$=6.4
t_4=7T/2=6.51	4.8 =ϕ_{4a}	4.0 =ϕ_{4b}	$\phi_4 = (_{34a}+\phi_{4b})/2$=4.4
t_5=9T/2=8.37	3.0 =ϕ_{5a}	2.4 =ϕ_{5b}	$\phi_5 = (\phi_{5a}+\phi_{5b})/2$=2.7
t_7=11T/2=10.23	2.2 =ϕ_{6a}	1.4 =ϕ_{6b}	$\phi_6 = (\phi_{6a}+\phi_{6b})/2$=1.8
t_7=13T/2=12.09	1.2 =ϕ_{7a}	0.8 =ϕ_{7b}	$\phi_7 = (\phi_{7a}+\phi_{7b})/2$=1.0
t_8=15T/2=13.95	0.8 =ϕ_{8a}	0.4 =ϕ_{8b}	$\phi_8 = (\phi_{8a}+\phi_{8b})/2$=0.6
t_9=17T/2=15.81	0.6 =ϕ_{9a}	0.2 =ϕ_{9b}	$\phi_9 = (\phi_{9a}+\phi_{9b})/2$=0.4

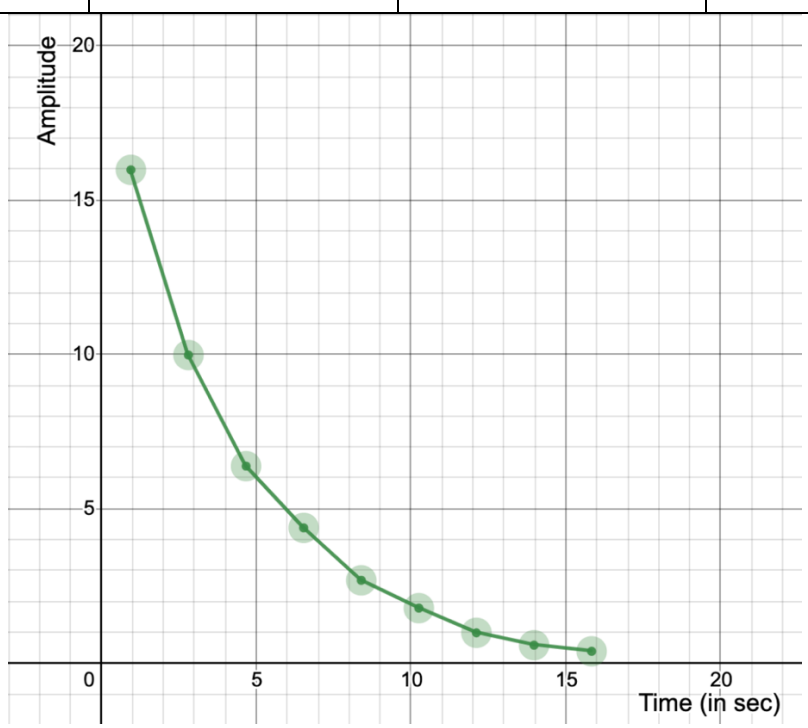

Table-7: Time period for i=0.6 amp.

Sl. No.	Time for 5 oscillations in sec	Time Period (T) T= **$(t_1+ t_2+ t_3)/15$ in sec**
1	**t_1= 8.88**	

2	**t_2= 9.32**	**1.81**
3	**t_3= 8.93**	

Table-8: Amplitude for i=0.6 amp.

Time (in sec)	Amplitude		$\phi_{avg} = (\phi_{ia}+\phi_{ib})/2$ i=1 to 5
	a	b	
t_1=T/2=0.90	12.2 =ϕ_{1a}	13.0 =ϕ_{1b}	$\phi_1 = (\phi_{1a}+\phi_{1b})/2$=12.6
t_2=3T/2=2.72	5.6 =ϕ_{2a}	5.8 =ϕ_{2b}	$\phi_2 = (\phi_{2a}+\phi_{2b})/2$=5.7
t_3=5T/2=4.52	2.6 =ϕ_{3a}	2.0 =ϕ_{3b}	$\phi_3 = (\phi_{3a}+\phi_{3b})/2$=2.3
t_4=7T/2=6.34	1.2 =ϕ_{4a}	0.8 =ϕ_{4b}	$\phi_4 = (_{34a}+\phi_{4b})/2$=1.0
t_5=9T/2=8.14	0.6 =ϕ_{5a}	0.2 =ϕ_{5b}	$\phi_5 = (\phi_{5a}+\phi_{5b})/2$=0.4

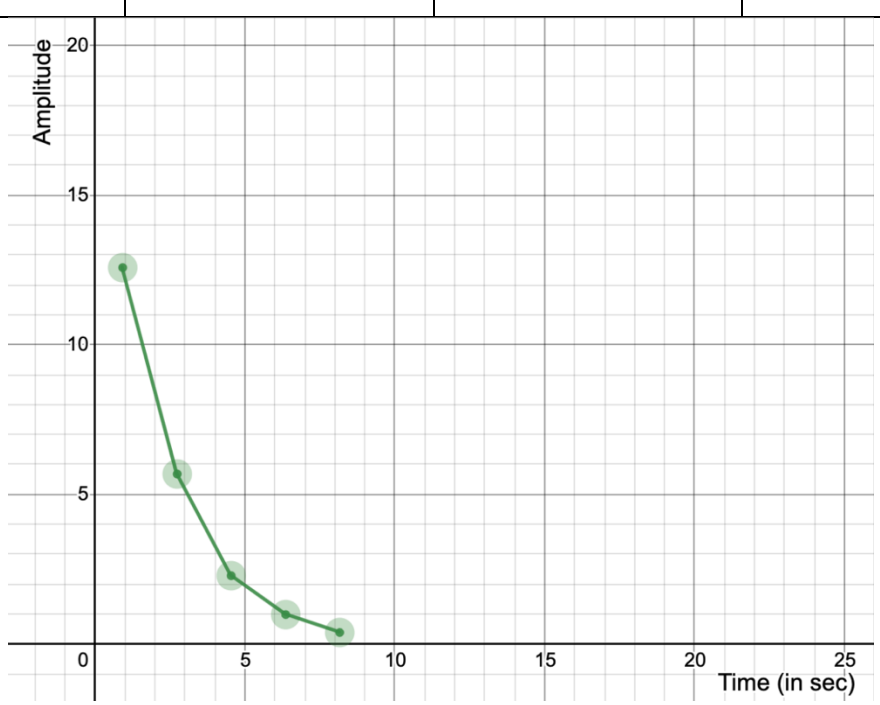

Table-9: Estimation of damping constant and resistance for different damping currents.

Damping Current	$ln\left(\frac{\phi_n}{\phi_{n+1}}\right)$	Avg. Logarithmic decrement λ	Time period T = $2\pi/\omega$ (in sec)	Damping constant $\beta = \frac{\lambda}{T}$	$\Omega_{res} = \sqrt{(\omega^2 - \beta^2)}$
0 A	0.0237	0.0289	1.88	0.0154 s^{-1}	3.3323 rad s^{-1}
	0.0304				
	0.0314				
	0.0324				
	0.0267				
0.2 A	0.1092	0.1221	1.87	0.0653 s^{-1}	3.3594 rad s^{-1}
	0.1226				
	0.1029				
	0.1284				
	0.1473				
0.4 A	0.4700	0.4370	1.86	0.2349 s^{-1}	3.3588 rad s^{-1}
	0.4463				
	0.3747				
	0.4884				
	0.4055				
0.6 A	0.7932	0.8625	1.81	0.4765 s^{-1}	3.4412 rad s^{-1}
	0.9076				
	0.8329				
	0.9163				

Table-10: Plot of frequency vs amplitude under forced oscillation (i = 0A)

Sl. No	Damping Current(A)	Time period in sec			Frequency (sec^{-1})	Amplitude			
		T_1	T_2	T_{avg}	f	ϕ_1	ϕ_1	ϕ_1	ϕ_{avg}
1	0 A	10.70	10.56	10.63	0.0941	0.6	0.6	0.8	0.667
2		4.50	4.42	4.46	0.2242	0.8	0.8	0.8	0.8
3		3.14	3.25	3.20	0.3130	1	0.8	1	0.933
4		2.20	2.38	2.29	0.4367	1.6	1.8	1.4	1.6
5		1.39	1.72	1.56	0.6431	2.4	2.2	1.6	2.066
6		1.90	1.70	1.80	0.5556	6.8	6.2	7.2	6.733
7		1.46	1.60	1.53	0.6536	1.4	0.8	1.2	1.133
8		1.37	1.39	1.38	0.7246	0.6	0.8	1	0.800
9		1.14	1.15	1.145	0.8734	1	0.6	0.6	0.733
10		0.92	1.03	0.975	1.0256	0.8	0.6	0.6	0.666

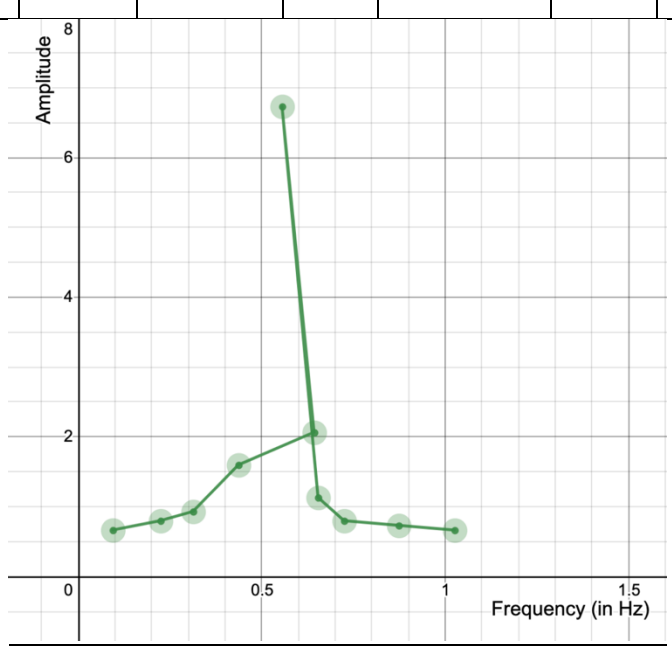

Table-11: Plot of frequency vs amplitude under forced oscillation (i = 0.2A)

Sl. No	Damping Current(A)	Time period in sec			Frequency (sec^{-1})	Amplitude			
		T_1	T_2	T_{avg}	f	ϕ_1	ϕ_1	ϕ_1	ϕ_{avg}
1	0.2 A	6.72	6.32	6.52	0.1534	0.8	0.8	0.6	0.733
2		7.74	4.62	6.18	0.1618	0.8	0.8	0.8	0.8
3		3.24	3.29	3.265	0.3063	1	1	0.8	0.933
4		2.83	2.79	2.81	0.3559	1.2	1	1.2	1.133
5		2.16	2.11	2.135	0.4684	3	3.4	3.4	3.266
6		2.05	1.91	1.98	0.5050	8	7.8	7.6	7.8
7		1.84	1.92	1.88	0.5319	7.4	7.6	7.4	7.466
8		1.46	1.40	1.43	0.6993	1.2	1.6	1.8	1.533
9		0.94	1.16	1.05	0.9523	0.8	0.6	0.6	0.666

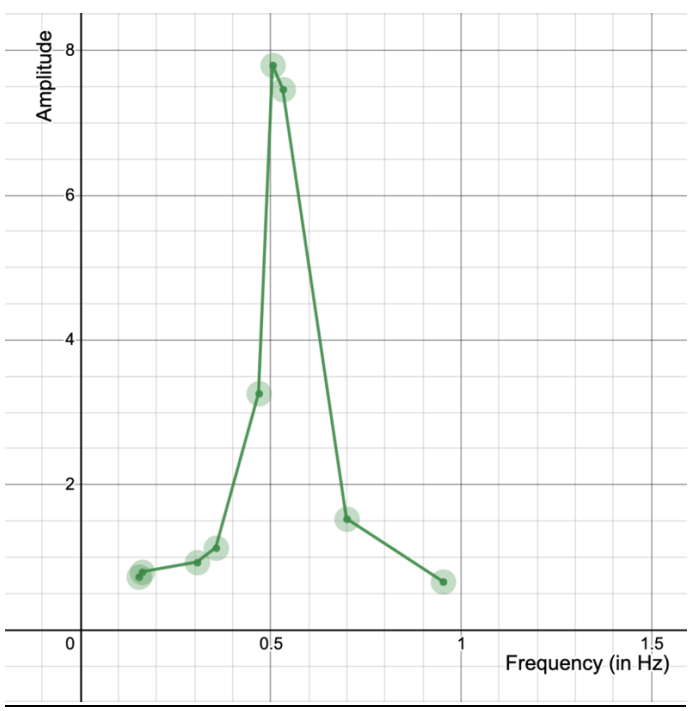

Table-12: Plot of frequency vs amplitude under forced oscillation (i = 0.4A)

Sl. No	Damping Current(A)	Time period in sec			Frequency (sec^{-1})	Amplitude			
		T_1	T_2	T_{avg}	f	ϕ_1	ϕ_1	ϕ_1	ϕ_{avg}
1	0.4 A	8.93	8.68	8.805	0.1136	0.8	0.6	0.6	0.666
2		8.49	8.51	8.72	0.1147	0.8	0.8	0.6	0.733
3		3.93	4.14	4.035	0.2478	0.8	0.8	0.8	0.8
4		3.01	2.94	2.975	0.3361	1	1	1	1
5		2.6	2.68	2.64	0.3788	1.2	1.2	1.4	1.266
6		2.14	2.29	2.215	0.4515	1.6	1.6	1.4	1.533
7		2.09	1.83	1.96	0.5102	2	2.2	2.2	2.133
8		1.81	1.72	1.765	0.5666	3.4	3.4	3.4	3.4
9		1.47	1.64	1.555	0.6431	1.8	2.2	2.0	2

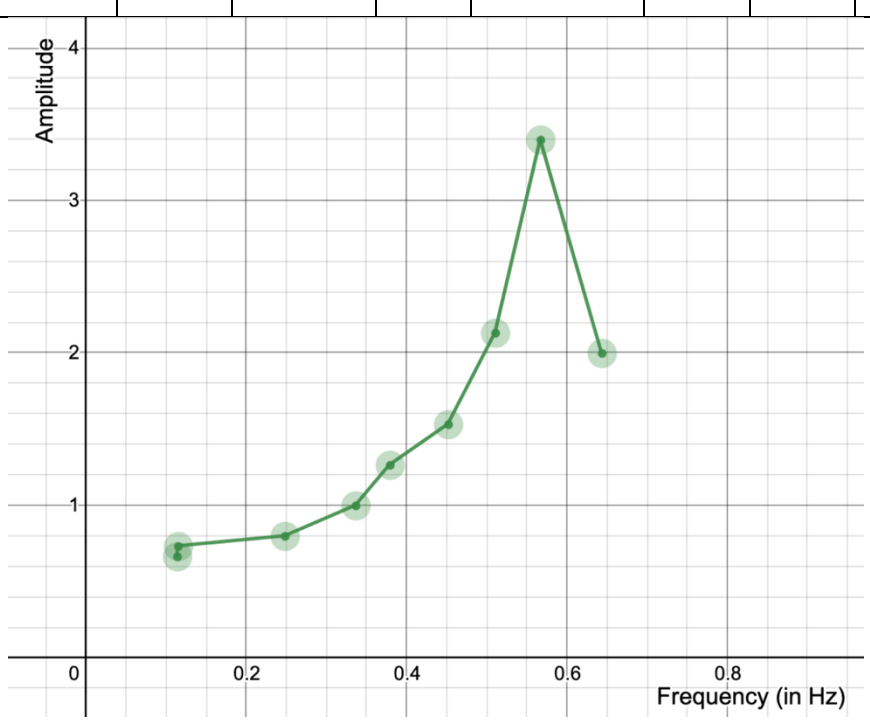

Table-13: Plot of frequency vs amplitude under forced oscillation (i = 0.6A)

Sl. No	Damping Current(A)	Time period in sec			Frequency (sec^{-1})	Amplitude			
		T_1	T_2	T_{avg}	f	ϕ_1	ϕ_1	ϕ_1	ϕ_{avg}
1	0.6 A	10.94	10.53	10.735	0.0932	0.6	0.6	0.8	0.666
2		3.86	3.94	3.9	0.2564	0.8	0.8	0.8	0.8
3		2.96	2.89	2.925	0.3419	1	0.8	1	0.933
4		2.39	2.04	2.215	0.4515	1.4	1.6	1.2	1.4
5		1.73	1.84	1.785	0.5602	1.8	2.0	2.0	1.933
6		1.32	1.40	1.36	0.7353	1.2	1.4	1.4	1.333
7		1.31	1.35	1.33	0.7519	0.8	0.8	1.0	0.866

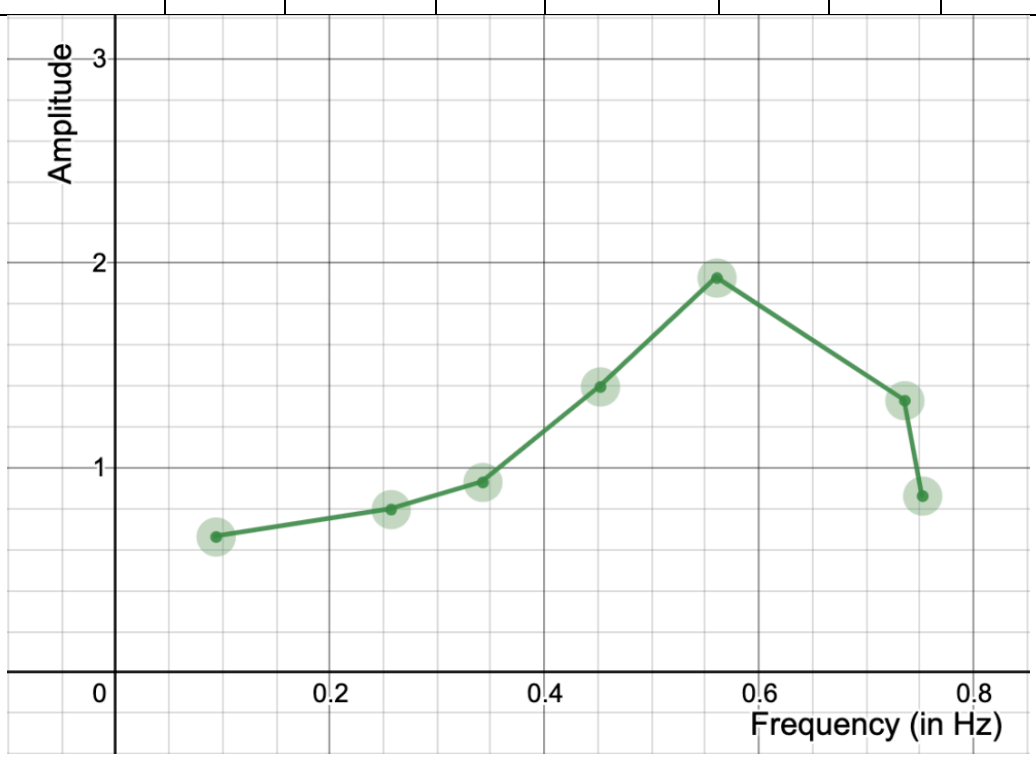

Table-14: Calculation of natural frequency

Damping Current	Logarithmic decrement λ	Damping const. β (Hz)	Resonance frequency (rad s^{-1})		Resonance amplitude ϕ_{res}	Natural angular Frequency from damped oscillation (rad s^{-1}) $\sqrt{\omega^2 + \beta^2)}$
			Observed ($2\pi f$)	Estimated (Ω_{res})		
0 A	**0.0289**	**0.0154**	**3.4906**	**3.3323**	**6.733**	**3.3324**
0.2 A	**0.1221**	**0.0653**	**3.1733**	**3.3594**	**7.8**	**3.3606**
0.4 A	**0.4370**	**0.2349**	**3.5599**	**3.3588**	**3.4**	**3.3953**
0.6 A	**0.8625**	**0.4765**	**3.5200**	**3.4412**	**1.933**	**3.5066**

Conclusions

1. Analyse the results and depict the interesting conclusion from the experiment.

   It can be observed that the transient solutions decay exponential over time. The amplitude of particular solution decreases with increase in β. At resonance , a maxima is observed in the amplitude vs frequency plot. Other findings are reasonably close and as expected to results found from theory. The estimated and predicted values of the resonance frequency are particularly close suggesting that our experiment is done properly. Determination of values of the damping constant, time period, and natural frequency shows the efficacy of the above procedure in measuring these parameters in a laboratory setting precisely.

2. Give any suggestions so that it can be incorporated to make the experiment simpler or more applicable practically.

   Optical/digital encoders can be used to accurately determine angular position and velocity or angular potentiometers can be used to accurately determine position.

Precautions

(i) While giving oscillation one should not give much pressure to the pulley.

(ii) While noting down the amplitudes during oscillations one should be very careful to avoid mistakes.

(iii) Do not disturb the spring while doing experiment.

(iv) Do not keep the pendulum in resonance condition without damping or with a very low damping for a long time.

(v) Observations should be taken after the pendulum gets stabilized.

Considerations

- What is the difference between simple pendulum and torsional pendulum? Is Pohl's pendulum a simple or a torsional one? Justify your answer.

  Ans : A simple pendulum is a pendulum in which a massive bob is suspended from a point and the restoring force is gravitational and proportional to the sine of the displacement, whereas in torsional pendulum the restoring force is torsional and proportional to the angle of twist. The Pohl's pendulum is a torsional pendulum, as the Pohl's wheel contains a spring which when twisted gives us the required oscillatory motion.

- Explain the terms under-damped, critically damped and over-damped oscillation. What is forced oscillation and resonance. How resonance does occur?

  Underdamped oscillation is characterised by transient solutions of the form of sinusoids with exponentially decaying amplitude.

  Mathematically the solution is of the form

$$Ae^{-\beta t}\cos(\sqrt{\omega_o^2-\beta^2}+\varphi)$$

  In case of critically damped oscillations, the pendulum swiftly returns to equilibrium without oscillating. Mathematically it is of the form

$$(At+B)e^{-\beta t}$$

  In case of over damped oscillation the pendulum slowly goes to equilibrium position asymptotically (creeping).

  Mathematically the transient solution is of the form

$$Ae^{(-\beta+\sqrt{\beta^2-\omega_o{}^2})t}+Be^{(-\beta-\sqrt{\beta^2-\omega_o{}^2})t}$$

  Oscillations are said to be forced when there is an external force or torque acting on the body undergoing motion. In this case we study the effect of a sinusoidal external force in which the steady state solution/particular solution is a sinusoid of the same angular frequency as the forcing function whose amplitude is a function of the driving frequency. When frequency of the forcing function is varied, there is a frequency where the amplitude is maximum. This phenomenon is termed as resonance and the particular frequency is called as the resonance frequency.

- What is logarithmic decrement factor? How is this related to nature of damping?

  Logarithmic decrement factor is the ratio of successive amplitudes in case of underdamped oscillations. It is denoted by λ. $\lambda=\beta T$ where β is the damping factor and T is the time period of the damped oscillations.